

Nhan (Nathan) Tran

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Education

- **Cornell University** *August 2022 - Current*
 - Ph.D., **Computer Science** (4.1/4.0 GPA). Minor in Cinematography. Adviser: [Abe Davis](#).
 - Research Interests: computational cinematography and photography, generative video, human-AI co-creation
- **Colorado School of Mines** *May 2020*
 - M.S. and B.S., **Computer Science + Robotics** (3.9/4.0 GPA). Adviser: [Tom Williams](#)
 - Masters Thesis: *Exploring Mixed Reality Robot Communication Under Different Types of Mental Workload*

Research experience

- **Google Research** *Research Intern, June 2026 - September 2026*
 - Researched new filmmaking workflows for generative reshooting models with [Neal Wadhwa](#) and [Stefan Stojanov](#).
 - Built a mobile capture system demonstrating this workflow: filmmakers plan camera movement in 3D space and time, capture with augmented reality guidance, and previsualize and manage hallucination risk before sending footage to the generative model.
 - Paper submitted to CHI 2027 [\[P.10\]](#) (patent pending).
- **Adobe** *Research Scientist Intern, May 2025 - November 2025*
 - Researched AI-assisted fine-cut editing with [Mackenzie Leake](#) and [Valentina Shin](#).
 - Formalized film idioms as audio-visual constraints, narrowing the search space for VLMs (e.g., Gemini), which excel at coarse scene understanding but lack frame-level temporal precision.
 - Built [MicroEdits](#), a plugin surfacing rhythm and view edits (L/J cuts, cut-on-action, cut-before-blink, reactions).
 - Evaluated with 16 professional editors, who reported less mechanical search and more creative exploration.
 - 🏆 First author, UIST 2026 **Best Paper Honorable Mention** (top 1%); patent pending [\[P.9\]](#).
- **Cornell University** *PhD Student Researcher, August 2022 - Current*
 - Researching technical HCI with [Prof. Abe Davis](#), spanning computational cinematography, photography, and augmented reality guidance; first author of 5 CHI/UIST papers (one award, one under review) [\[P.6–P.10\]](#).
 - Built [CineCraft](#), a mobile app unifying shot planning, AR-guided capture, and post-processing (CHI 2026) [\[P.8\]](#).
 - Built [ARticate](#), AR guidance that learns rotational frames from user demonstrations (CHI 2025) [\[P.7\]](#).
 - Built Personal Time-Lapse ([MeCapture](#)), an AR app using 3D tracking and computational imaging to capture long-term body changes for telehealth (e.g., wound healing, recovery); users in 14 countries (UIST 2024) [\[P.6\]](#).
 - Co-wrote a grant proposal with Prof. Davis (PI) on smartphone-based ophthalmic imaging with AI for early detection of eye disease; [awarded \\$75K](#), with a clinical collaboration underway.
- **Robust AI** *Research Intern, June 2020 - September 2020*
 - Researched augmented reality and human-robot interaction with [Alex Trevor](#), [Crystal Chao](#), and [Rodney Brooks](#).
 - Built Unity3D simulator and mixed reality visualizer (iOS and HoloLens 2) with ROS integration, enabling remote robot control and real-time debugging for virtual robot path planning in real environments.
- **Mines Interactive Robotics Research Lab** *Research Assistant and Project Lead, 2017 - 2020*
 - Researched AR/VR and human-robot interaction with [Prof. Tom Williams](#); HRI/THRI papers [\[P.1–P.5\]](#).

Publications

ACM CHI, UIST, HRI are the premier venues for technical Human-Computer Interaction (HCI) publications, which are fully peer-reviewed and have an acceptance rate of 20-25%.

[\[P.10\]](#) **Nhan Tran** et al. *A paper on AI-assisted cinematography with generative video*. Under submission at the ACM CHI Conference on Human Factors in Computing Systems (CHI 2027).

- [P.9] **Nhan Tran**, Hijung Valentina Shin, Abe Davis, Mackenzie Leake. *Micro-Edits: Computational Support for Fine-Scale Edits in Film*. 39th Annual ACM Symposium on User Interface Software and Technology (UIST 2026).  **Best Paper Honorable Mention (top 1%)**. microedits.github.io
- [P.8] **Nhan Tran**, Sam Belliveau, Zixin Xu, Abe Davis. *CineCraft: Unified Shot Planning, Capture, and Post-Processing for Mobile Cinematography*. ACM CHI Conference on Human Factors in Computing Systems (CHI 2026). trannhan.com/cinecraft/
- [P.7] **Nhan Tran**, Ethan Yang, Abe Davis. *Articulate: Interactive Visual Guidance for Demonstrated Rotational Degrees of Freedom in Mobile AR*. ACM CHI Conference on Human Factors in Computing Systems (CHI 2025). trannhan.com/articulate/
- [P.6] **Nhan Tran**, Ethan Yang, Angelique Taylor, Abe Davis. *Personal Time-Lapse*. 37th Annual ACM Symposium on User Interface Software and Technology. (UIST 2024). [MeCapture.com](https://mecapture.com)
- Free iOS app with potential applications in telehealth for documenting long-term bodily changes. Users in 14 countries. Featured in [press coverage](#).
- [P.5] **Nhan Tran**, Trevor Grant, Thao Phung, Leanne Hirshfield, Christopher Wickens, Tom Williams. *Now Look Here! Mixed Reality Improves Robot Communication Without Cognitive Overload*. 25th International Conference on Human-Computer Interaction (HCII 2023).
- [P.4] Landon Brown, Jared Hamilton, Zhao Han, Albert Phan, Thao Phung, Eric Hansen, **Nhan Tran**, Tom Williams. *Best of Both Worlds? Combining Different Forms of Mixed Reality Deictic Gestures*. ACM Transactions on Human-Robot Interaction (THRI 2022).
- [P.3] Jared Hamilton and Thao Phung and **Nhan Tran** and Tom Williams. *What's The Point? Tradeoffs Between Effectiveness and Social Perception When Using Mixed Reality to Enhance Gesturally Limited Robots*. 16th ACM/IEEE International Conference on Human-Robot Interaction (HRI 2021).
- [P.2] Tom Williams and Matthew Bussing and Sebastian Cabrol and Elizabeth Boyle and **Nhan Tran**. *Mixed Reality Deictic Gesture for Multi-Modal Robot Communication*. 14th ACM IEEE International Conference on Human Robot Interaction (HRI 2019)
- [P.1] Tom Williams, **Nhan Tran**, Josh Rands, Neil T. Dantam . *Augmented, Mixed, and Virtual Reality Enabling of Robot Deixis*. 10th International Conference on Virtual, Augmented and Mixed Reality (VAMR 2018).

Invited Talks and Presentations

- Oral presentation of *Micro-Edits* at the [UIST 2026](#) conference in Detroit, MI (upcoming) November 2026
- Selected for the UIST 2026 Doctoral Symposium (small cohort presenting dissertation research) November 2026
- Oral presentation at the CHI 2026 conference April 2026
- Invited speaker at Columbia University's HCI Seminar November 2025
- Live demo "MeCapture" app at [SIGGRAPH 2025](#) conference August 2025
- Oral presentation at the [CHI 2025](#) conference May 2025
- Finalist at Cornell's Annual [Eastman-Rice Persuasive Debate Contest](#). Won 2nd place. April 2025
- Invited Lightning Talk speaker at [New York City Computer Vision Day](#) February 2025
- Invited speaker at Cornell Tech [XR Monthly Seminar](#) (Faculty Host: Harald Haraldsson) January 2025
- Invited speaker at Princeton [Vision & Graphics Seminar](#) (Faculty Host: Adam Finkelstein) November 2024
- Finalist at Three Minute Thesis (3MT) Competition, Harvard Engineering's [Ivy Symposium](#) October 2024
- Oral presentation at [UIST 2024](#) conference October 2024
- Oral presentation at [HCII 2023](#) conference August 2023

Work experience (selected)

- **Robust AI** *Software Engineer, Robot Perception, June 2020 - August 2022*
 - Developed robot perception systems (obstacle avoidance, object detection, monocular depth estimation model).
 - Built simulation and visualization tools that allow rapid iteration of robotic development and sensor inspection.
- **Facebook** *Production Engineer Intern, May 2019 - August 2019*
 - Saved team from capacity overloads by building automation tools that analyze historical data and control the sampling rates of machine learning experiments. Received a return full-time offer.
- **NASA-Caltech Jet Propulsion Laboratory** *Software Intern (Web/Full-stack), June 2018 - August 2018*
 - Researched and resolved bottlenecks in Mars 2020 Rover's Sampling and Caching Subsystem's cleaning process.
 - Reduced 50% lead time by designing an assembly tracking system to optimize operations workflows.
- **Google Nest** *Software Engineering Intern, May 2017 - August 2017*
 - Reduced man-hours by 90% by developing a visualization tool for internal robots to automate product testing.

Service

- Reviewer, ACM CHI Conference on Human Factors in Computing Systems (2025, 2026, 2027)
- Reviewer, ACM SIGGRAPH (2026); Poster Reviewer, ACM SIGGRAPH (2025)
- Mentor and Panelist, Cornell Bowers CIS undergraduate research program (BURE 2023, 2024)
- Student Volunteer, Cornell Computer Science Visit Days (2023, 2024, 2025)
- Reviewer, ACM/IEEE International Conference on Human-Robot Interaction (2021, 2022)
- Web chair, ACM/IEEE International Conference on Human-Robot Interaction (2021)
- Publicity chair, Human-Robot Interaction Pioneers Workshop at HRI 2021 (2021)
- Scholarship reviewer, Greenhouse Scholars (2020)
- Student Outreach Workshop organizer, AAAI Conference on Artificial Intelligence (2018, 2019)

Teaching experience

- **Cornell University** *Graduate Teaching Assistant, Fall 2022, Spring 2023*
 - Spring 2023: [CS4670 Computer Vision](#) with 350+ students.
 - Fall 2022: [CS2110 Data Structures](#), one of Cornell's largest computer science courses with 700+ students.

Filmography

Beyond my research, I've cultivated filmmaking skills through Cornell's Cinematography program, serving in various roles from director to cinematographer. These creative experiences directly inform my research on improving tools and workflows for content creators. Student productions:

- [Für Elise](#) (Short Film), self-directed Writer and Director, 2024
- Earworm (Short Film) directed by Jake Walter First Assistant Camera, 2024
- Silence (Music Video) directed by Katherine Xu First Assistant Camera, 2024
- 50/50 (Short Film) directed by Justin Lee Director of Photography, 2024
- Premeditated (48 Hour Filmmaking competition) directed by Katherine Xu Lighting/Gaffer, 2024
- [The Tiny Explorer](#) (Short Film), co-directed with Waki Kamino and Peter Wu Actor and Co-Director, 2023
- [My Robot](#) (Short Film), self-directed Writer and Director, 2023
- Facade (Short Film) directed by Krisha Jivani Director of Photography, 2023
- [Class Crush](#) (Music Video) directed by Katherine Xu Production Assistant, 2023